

LABORATORY RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 4

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TITLE

Implementation of a Student Marks Calculator using HTML, CSS, and JavaScript

AIM

To design and implement an interactive web-based calculator to compute total marks, average, and grade using conditional logic in JavaScript.

OBJECTIVES

- Create a structured HTML interface with input fields for subject marks.
- Apply CSS styling for a clean, centered calculator interface.
- Implement a JavaScript function to retrieve DOM values and perform arithmetic calculations.
- Use conditional statements (if-else) to evaluate grades.
- Dynamically update the DOM with calculated results.

TASK DESCRIPTION

Design an interactive web-based "Student Marks Calculator". The application must feature input fields for three subjects and a submission button. Upon clicking the button, a JavaScript function should extract the input values, calculate the total sum and average, determine the final grade based on the average, and display these metrics dynamically on the webpage.

ALGORITHM/PROCEDURE

1. Create an HTML structure containing a container div, three numeric input fields, a button, and a result display div.
2. Apply CSS styles to center the container, add borders, padding, and round the corners for better UI.
3. Define a JavaScript function calculate() that is triggered by the button's onclick event.
4. Inside the function, fetch the values from the input fields using document.getElementById().value and cast them to numbers.
5. Compute the total sum of the three subjects and determine the average.
6. Utilize an if-else ladder to assign a grade (A+, A, B, C, or Fail) based on the calculated average.
7. Update the innerHTML of the result div to output the total, average (using toFixed(2)), and grade.

PROGRAM/SOURCE CODE

The following code represents the complete HTML, CSS, and JavaScript implementation with detailed explanatory comments:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Student Marks Calculator</title>
  <style>
    /* Base page styling */
    body {
      font-family: Arial, sans-serif;
      text-align: center;
      margin-top: 50px;
      background-color: #f9f9f9;
    }

    /* Container card to center and frame the calculator */
    .container {
      width: 300px;
      margin: auto;
      padding: 20px;
      border: 2px solid black;
      border-radius: 10px;
      background-color: #ffffff;
      box-shadow: 0 4px 8px rgba(0,0,0,0.1);
    }
  </style>
</head>
<body>
  <div class="container">
    <div>
      <input type="text"/>
      <input type="text"/>
      <input type="text"/>
      <button>Calculate</button>
    </div>
    <div>
      <div><div></div></div>
      <div></div>
    </div>
  </div>
</body>
</html>
```

```

/* Input fields layout and spacing */
input {
    width: 80%;
    padding: 8px;
    margin: 6px 0;
    border: 1px solid #ccc;
    border-radius: 4px;
    font-size: 14px;
}

/* Submit button styling */
button {
    padding: 10px 20px;
    cursor: pointer;
    margin-top: 10px;
    background-color: #007bff;
    color: white;
    border: none;
    border-radius: 4px;
    font-weight: bold;
}

button:hover {
    background-color: #0056b3;
}

/* Output text container styling */
#result {
    margin-top: 15px;
    font-weight: bold;
    font-size: 15px;
    line-height: 1.6;
}
</style>
</head>
<body>

<!-- Calculator Container -->
<div class="container">
    <h2>Student Marks Calculator</h2>

    <!-- Input fields to capture marks for three individual subjects -->
    <input type="number" id="m1" placeholder="Subject 1 Marks"><br>
    <input type="number" id="m2" placeholder="Subject 2 Marks"><br>
    <input type="number" id="m3" placeholder="Subject 3 Marks"><br>

    <!-- Action button bound to the calculate() JavaScript function -->
    <button onclick="calculate()">Calculate</button>

    <!-- Target container where calculated metrics will be rendered -->
    <div id="result"></div>

```

```

</div>

<script>
  // Function triggered when the user clicks the "Calculate" button
  function calculate() {
    // 1. Retrieve the input values and convert them from Strings
to Numbers
    let m1 = Number(document.getElementById("m1").value);
    let m2 = Number(document.getElementById("m2").value);
    let m3 = Number(document.getElementById("m3").value);

    // 2. Perform arithmetic operations to calculate total and
average
    let total = m1 + m2 + m3;
    let average = total / 3;
    let grade = "";

    // 3. Conditional ladder to evaluate performance and assign
letter grade
    if (average >= 90) {
      grade = "A+";
    } else if (average >= 75) {
      grade = "A";
    } else if (average >= 60) {
      grade = "B";
    } else if (average >= 40) {
      grade = "C";
    } else {
      grade = "Fail";
    }

    // 4. Inject the calculated values dynamically into the DOM
    // toFixed(2) rounds the average to 2 decimal places
    document.getElementById("result").innerHTML =
      "Total: " + total + "<br>" +
      "Average: " + average.toFixed(2) + "<br>" +
      "Grade: " + grade;
  }
</script>
</body>
</html>

```

CODE EXPLANATION

Element / Method	Purpose & Explanation
document.getElementById()	Retrieves an HTML element by its unique ID to read user input or update content dynamically.
Number()	Converts the retrieved string value from

	the input field into a numeric datatype for calculations.
onclick	An event attribute attached to the button that executes the specific calculate() function when clicked.
if-else if	Conditional statements used to evaluate the computed average and assign the corresponding grade string.
innerHTML	A property used to dynamically inject calculated results and text directly into the webpage document.
toFixed(2)	Formats the resulting average to consistently display exactly two decimal places.

TEST CASES AND EXECUTION DATA

Test Case	Verification	Expected Result	Observed Result	Status
TC-01	Valid inputs calculation	Inputs 89, 92, 87 -> Total 268, Avg 89.33, Grade A	Result displayed correctly as per conditions	Pass
TC-02	High tier average	Inputs 95, 92, 98 -> Average ≥ 90	Grade A+ evaluated and displayed correctly	Pass
TC-03	Failing tier average	Inputs 30, 20, 35 -> Average < 40	Grade Fail evaluated and displayed correctly	Pass
TC-04	Decimal formatting	Inputs result in recurring decimal	toFixed(2) limits to exactly two decimal places	Pass

OUTPUT

The following browser output screenshots depict the execution of the 4 Test Cases with clear white backgrounds for printing:

TC-01: Valid inputs calculation (Marks: 89, 92, 87 -> Grade A)

Student Marks Calculator

89

▲▼

92

▲▼

87

▲▼

Calculate

Total: 268

Average: 89.33

Grade: A

TC-02: High tier average (Marks: 95, 92, 98 -> Grade A+)

Student Marks Calculator

95

▲▼

92

▲▼

98

▲▼

Calculate

Total: 285

Average: 95.00

Grade: A+

TC-03: Failing tier average (Marks: 30, 20, 35 -> Grade Fail)

Student Marks Calculator

30

20

35

Calculate

Total: 85

Average: 28.33

Grade: Fail

TC-04: Decimal formatting check (Marks: 70, 75, 72 -> Avg: 72.33, Grade B)

Student Marks Calculator

70

75

72

Calculate

Total: 217

Average: 72.33

Grade: B

RESULT

Thus, the interactive Student Marks Calculator was successfully designed and implemented using HTML, CSS, and JavaScript. The web interface correctly captures user inputs, performs arithmetic and logical evaluations, and dynamically outputs the total, average, and grade securely without requiring page reloads.